

Topic 2.12 - Logarithmic Function Manipulation

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Expand and condense logarithmic expressions using product, quotient, and power properties.
- Use change of base and natural logarithms to evaluate and compare logarithms.
- Explain algebraic and graphical equivalences while preserving domain restrictions.

Section A: Foundation and representation

Question 1

Core skill

No calculator

Expand $\log_5(125x^4y)$ for $x > 0, y > 0$.

Question 2

Core skill

No calculator

Expand $\ln\left(\frac{e^3(x+1)^2}{\sqrt{x}}\right)$ for $x > 0$.

Question 3

Core skill

No calculator

Expand $\log_3 \left(\frac{(x-2)^6}{9(x+4)} \right)$ and state the original domain.

Question 4

Core skill

No calculator

Condense $3\log_2 x + \log_2(x-1) - 2\log_2(x+3)$ and state the domain.

Question 5

Core skill

No calculator

Condense $\frac{1}{2} \ln a + \frac{3}{2} \ln b - \ln c$ for positive a, b, c .

Question 6

Core skill

Calculator permitted

Use change of base to approximate $\log_{12} 500$ to four decimal places.

Section B: Application, comparison, and error analysis

Question 7

Proof

No calculator

Prove that $\log_{1/3} x = -\log_3 x$ for $x > 0$.

Question 8

Domain reasoning

No calculator

Show that $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ are equivalent only on their shared domain. Compare their domains.

Question 9

Graph equivalence

No calculator

Describe the graph relationship between $f(x) = \log_5(25x)$ and $g(x) = \log_5 x$.

Question 10

Error analysis

No calculator

A student condenses $\ln(x - 2) + \ln(x + 2)$ to $\ln(x^2 - 4)$ and changes the domain to $x < -2$ or $x > 2$. Explain the domain error.

Question 11

Core skill

No calculator

Solve for k if $\log_2(8x) = \log_2 x + k$ for all $x > 0$.

Question 12

Core skill

No calculator

Rewrite $4\log_3 x - 2\log_3 y + \log_3 9$ as a single logarithm.

Section C: AP-style multiple choice**MCQ 1**

No calculator

Which is equivalent to $\log_b(MN^2)$ for positive M, N ?

- A. $\log_b M + 2\log_b N$
- B. $\log_b M \cdot 2\log_b N$
- C. $\log_b(M + N^2)$
- D. $2\log_b(MN)$

MCQ 2

No calculator

Which expression condenses to $\log(x^3/y)$?

- A. $3\log x - \log y$
- B. $\log(3x) - \log y$
- C. $\log x - 3\log y$
- D. $\log(x/y^3)$

MCQ 3

No calculator

Which equality is false in general?

- A. $\log_b(xy) = \log_b x + \log_b y$
- B. $\log_b(x/y) = \log_b x - \log_b y$
- C. $\log_b(x + y) = \log_b x + \log_b y$
- D. $\log_b(x^p) = p \log_b x$ on the shared domain

MCQ 4

No calculator

$\log_{25} x$ is equal to:

- A. $2 \log_5 x$
- B. $\frac{1}{2} \log_5 x$
- C. $\log_5 x + 2$
- D. $\log_5(x/2)$

Section D: Free-response reasoning

FRQ 1 - Expand, condense, and preserve domain

No calculator

6 points

Let $E(x) = 2\ln(x - 1) + \frac{1}{2}\ln(x + 3) - \ln 4$. (a) State the domain of E . (b) Condense E into one logarithm. (c) Expand $\ln\left(\frac{(x - 1)^2\sqrt{x + 3}}{4}\right)$ and explain why the domain must remain the same as in part (a). (d) Evaluate $E(5)$ exactly.

FRQ 2 - Change of base and graphical meaning

No calculator

6 points

Define $f(x) = \log_2(16x)$ and $g(x) = \log_8 x$ for $x > 0$. (a) Rewrite each function using $\log_2 x$. (b) Describe the graph transformation from $y = \log_2 x$ to each function. (c) Solve $f(x) = g(x)$. (d) Find the common output.
